

# **UNIT 1**

## **INTRODUCTION TO MACHINE LEARNING**

Comprehensive Exam Notes

*MCA 521-4 - Machine Learning*

*Mid-Semester Examination Preparation*

Pattern: 5 Questions (3 Compulsory + 2 Optional)

Total Marks: 50

Syllabus Coverage: 2.5 Units

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## SECTION A: INTRODUCTION TO ARTIFICIAL INTELLIGENCE

(Expected Marks: 5-8)

### 1.1 Definition of Artificial Intelligence

AI aims to create hardware and software systems that exhibit thinking comparable to humans.

- Term coined by John McCarthy in 1956 at the Dartmouth Conference
- Concerned with the design of intelligence in an artificial device
- Goal: Not just to understand intelligence, but to BUILD intelligent entities

### 1.2 What is Intelligence?

Intelligence is the cognitive ability to:

- Learn from experience and reason well
- Remember important information and provide basis for judgement
- Gain knowledge by studying, practicing, or experiencing
- Acquire, interpret, select, and organize sensory information
- Arrive at desired solutions from present situations
- Use, comprehend, speak, and write language

### 1.3 Intelligence vs Wisdom

| Intelligence          | Wisdom                     |
|-----------------------|----------------------------|
| Think logically       | Grasp human nature         |
| Conceptualize reality | Understand paradox         |
| Problem-solving       | Experiential understanding |

### 1.4 Four Definitions of AI (IMPORTANT)

**NOTE:**

This is a very common 2-mark question!

- Thinking Humanly: Thinking like a person (cognitive modeling)
- Thinking Rationally: Modeling thinking as logical process (laws of thought)
- Acting Humanly: Acting like a person (Turing Test approach)
- Acting Rationally: Performing actions that increase value of agent state

### 1.5 The Turing Test (IMPORTANT - 3-5 Marks)

**NOTE:**

Proposed by Alan Turing (1950). If an interrogator cannot reliably distinguish between a human and a computer based on replies to questions, the computer possesses intelligence.

Requirements for passing the Turing Test:

- Natural Language Processing (NLP)
- Knowledge Representation
- Automated Reasoning
- Machine Learning

### 1.6 Foundations of AI

AI draws from: Philosophy, Mathematics, Economics, Neuroscience, Psychology, Computer Engineering, Control Theory, and Linguistics.

## 1.7 Applications of AI

- Computer Vision & Image Recognition
- Robotics & Autonomous Navigation
- NLP, NLU, and Speech Processing
- Expert Systems & Medical Diagnosis
- Autonomous Vehicles & Game Playing

## SECTION B: KNOWLEDGE REPRESENTATION

(Expected Marks: 5-8)

### 2.1 Definition and Importance

Knowledge representation is a fundamental requirement for AI systems. They must be able to:

- Identify knowledge
- Acquire knowledge
- Store knowledge
- Process knowledge
- Produce intelligent results

**NOTE:**

Key Principle: A natural and effective representation is often as important as the algorithm itself.

### 2.2 Types of Knowledge Representation

#### A. Classical Symbol-Based Representations

Traditional AI uses symbols to encode rules and relationships. Based on formal logic and rule-based systems. Examples: Predicate logic, semantic networks, frames.

- Explicit representation of knowledge
- Human-readable rules
- Good for structured domains

#### B. Visual and Causal Representations

Modern analytical AI uses visual frameworks to represent complex relationships. Causal models show cause-effect relationships. Used in Bayesian networks and decision trees.

#### C. Knowledge Representation in Machine Learning

Shift from hand-coded rules to data-driven structures. Representation learned automatically from data. Examples: Neural network weights, embeddings, feature vectors.

### 2.3 Components

- Syntax: Rules for constructing valid expressions
- Semantics: Meaning of expressions
- Inference Procedures: Methods for deriving new knowledge

## SECTION C: DATA - TYPES, LEVELS, AND PROCESSING

(Expected Marks: 8-10)

### 3.1 Definition of Data

Data is the fundamental building block for AI/ML systems. Facts and figures collected, analyzed, and summarized for presentation and interpretation.

**NOTE:**

Key Argument: Quality and nature of data are MORE IMPORTANT than the specific algorithm.

### 3.2 Dataset Structure

| Component | Description            | Example            |
|-----------|------------------------|--------------------|
| Samples   | Each row = one entity  | One customer       |
| Features  | Each column = property | Age, gender, price |
| Labels    | Feature to predict     | Survived (1/0)     |

### 3.3 Two Big Categories of Data

#### A. Metric (Quantitative) Data

Represents amount or degree. Numbers you can do math with. Examples: Age, weight, income, height, temperature. Further classified into Discrete and Continuous.

#### B. Nonmetric (Qualitative/Categorical) Data

Represents categories or labels. Describes differences in type or kind. Examples: Occupation, marital status, color, gender. Further classified into Nominal and Ordinal.

### 3.4 Levels of Data (VERY IMPORTANT)

**NOTE:**

This is a very common 5-mark question! Memorize definitions and examples.

#### 1. NOMINAL DATA

Labels variables WITHOUT any order or quantitative value.

- Cannot do numerical tasks or give order to sort data
- Only MODE is valid measure of central tendency
- Examples: Hair color, Marital status, Nationality, Gender, Eye color
- ML Use: One-hot encoding for Decision Trees, Logistic Regression
- Origin: "Nominal" from Latin "nomen" = name

#### 2. ORDINAL DATA

Natural ordering present, but intervals between values NOT equal.

- Can rank/order data but cannot assume equal spacing
- MODE and MEDIAN valid; MEAN is NOT valid
- Examples: Customer satisfaction (1-10), Letter grades (A,B,C), Rankings, Economic status
- ML Use: Sentiment analysis (Positive > Neutral > Negative)
- Key: Numbers indicate relative positions only

#### 3. INTERVAL DATA

Numerical scale with EQUAL intervals but NO TRUE ZERO.

- Can add/subtract but NOT multiply/divide meaningfully
- No true zero point

- Examples: Temperature (Celsius/Fahrenheit), IQ scores, Calendar years, Likert scales
- Important: Cannot say "80C is twice as hot as 40C"
- Tests: t-tests, ANOVA, Pearson correlation

### 4. RATIO DATA

Like interval data but WITH TRUE ZERO.

- Enables meaningful ratios
- All mathematical operations valid
- Examples: Height, Weight, Income, Age, Time duration, Number of sales
- Important: Can say "80,000 salary is twice 40,000"
- Tests: All parametric tests, coefficient of variation

### 3.5 Data Level Operations Summary

| Level    | Order | Equal Int. | True Zero | Mode | Median | Mean | Ratio |
|----------|-------|------------|-----------|------|--------|------|-------|
| Nominal  | No    | No         | No        | Yes  | No     | No   | No    |
| Ordinal  | Yes   | No         | No        | Yes  | Yes    | No   | No    |
| Interval | Yes   | Yes        | No        | Yes  | Yes    | Yes  | No    |
| Ratio    | Yes   | Yes        | Yes       | Yes  | Yes    | Yes  | Yes   |

### 3.6 Quantitative Data Classification

#### A. DISCRETE DATA

Distinct or separate values. Whole numbers only. Cannot be broken into decimals. Countable and finite.

- Examples: Number of students, customers, employees, players
- Key: You cannot have 5.7 customers
- Representation: Bar graph, number line, frequency table

#### B. CONTINUOUS DATA

Fractional numbers. Can be divided into smaller levels. Can take ANY value within a range.

- Examples: Height, speed, time taken, market share price, temperature
- Key: 4 minutes 32.6 seconds is valid



## SECTION D: EXPLORATORY DATA ANALYSIS (EDA)

(Expected Marks: 10-12)

### 4.1 Definition of EDA

EDA is an essential step in data analysis that focuses on understanding patterns, relationships, and distributions within a dataset using statistical methods and visualizations.

- Python Libraries: pandas, NumPy, plotly, matplotlib, seaborn

### 4.2 Common EDA Techniques

- Data Inspection: Check size, organization, data types, basic summary
- Handling Missing/Duplicate Data: Find and fix empty values/repeated rows
- Univariate Analysis: Study one variable at a time (distribution, trend, outliers)
- Bivariate Analysis: Compare two variables (relationship)
- Multivariate Analysis: Analyze three+ variables together (deeper patterns)

### 4.3 7 Steps in EDA (VERY IMPORTANT)

#### NOTE:

Common 5-mark question! Memorize all 7 steps with brief descriptions.

- 1. Defining Questions/Hypothesis: Identify objectives. Example: "Why are we losing customers?"
- 2. Data Collection: Quality affects results. Categories: Primary, Secondary, Third-party (Big Data)
- 3. Cleaning Data: Remove errors, duplicates, outliers. Fix typos. Handle missing values.
- 4. Data Visualization: Use charts and graphs to understand patterns.
- 5. Feature Engineering: Use domain knowledge to extract useful features.
- 6. Selection of Analysis Technique: Choose appropriate statistical/ML methods.
- 7. Summarizing Results: Draw conclusions and actionable insights.

### 4.4 EDA in Python - Key Functions

#### Importing Libraries

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import warnings as wr
wr.filterwarnings('ignore')
```

#### Key Analysis Functions

- `df.shape()`: Returns (rows, columns) - overview of dataset size
- `df.info()`: Shows records, data types, missing values, memory usage
- `df.describe().T`: Statistical summary (count, mean, std, min, quartiles, max)
- `df.columns.tolist()`: List of column names for easy manipulation
- `df.isnull().sum()`: Total null values per column - identifies data gaps

### 4.5 Why Process/Clean Data?

- Raw data is often "dirty" with errors or gaps
- Data examination is an "investment in multivariate insurance"
- Common issues: Insufficient, poor quality, incorrect, missing, irrelevant, duplicates
- Primary tasks: Data Cleaning + Feature Management

### 4.6 Dimensions of Data Quality (8 Dimensions)

- 1. Completeness: No missing values
- 2. Accuracy: Values reflect reality
- 3. Consistency: Uniformity across sources
- 4. Validity: Compliance with required formats
- 5. Currency: Data is up to date
- 6. Accessibility: Ease of retrieval
- 7. Reliability: Trustworthiness of source
- 8. Usability: Interpretability for end-user

## SECTION E: MISSING DATA & IMPUTATION TECHNIQUES

(Expected Marks: 8-10)

### 5.1 Why Handle Missing Values?

- Missing values reduce accuracy and create bias
- Break algorithms requiring complete data
- Benefits: Keeps models accurate, prevents bias, maintains sample size

### 5.2 Reasons Behind Missing Values

- Technical issues: Failed data collection, transmission errors
- Human errors: Incorrect data entry, processing oversights
- Privacy concerns: Sensitive information withheld
- Data processing issues: Errors during transformation

### 5.3 Types of Missing Data (VERY IMPORTANT)

#### NOTE:

This is a very common 5-mark question! Understand the differences clearly.

#### A. MCAR - Missing Completely at Random

Missingness occurs randomly, NOT related to ANY variable.

- Purely haphazard, no systematic pattern
- Test: Compare data with/without missing values using t-test
- If no difference in means -> MCAR
- Examples: Scale ran out of batteries, random population sample
- Handling: Listwise deletion acceptable if large sample

#### B. MAR - Missing at Random

Missingness depends on OTHER observed variables, NOT on missing value itself.

- Missing randomly within sub-samples
- Example: GPA missing for students in SPECIFIC schools (not all)
- If missing across ALL schools -> MCAR
- If missing for specific schools -> MAR

#### C. NMAR (MNAR) - Not Missing at Random

Missingness is DIRECTLY related to the missing value itself.

- Has structure/pattern, cannot treat as random
- Often undetectable and problematic
- Examples: High-income individuals not reporting income
- All students from specific schools missing data

### 5.4 Missing Data Handling Techniques

#### A. Complete Removal Methods

- Listwise Deletion: Omit entire row if ANY variable missing. Use when MCAR and large sample.
- Pairwise Deletion: Use all available data for each procedure. Different sample sizes for different analyses.

#### B. Univariate Imputation - Numerical

- Mean Imputation: Replace with mean. Simple but reduces variance. Best for MCAR, small proportion missing.
- Median Imputation: More robust to outliers than mean. Use for skewed distributions.
- Random Sample Imputation: Replace with random observation. Preserves mean and std. Use for MCAR, >10% missing.

- End of Distribution: Replace with value at distribution end. Captures informative missingness. Use for NMAR.
- Arbitrary Value: Assign value far from range (e.g., 999). Creates indicator that value was missing.

### C. Univariate Imputation - Categorical

- Mode Imputation: Replace with most frequent category. Use when MAR and missing < 5%.
- Missing as Category: Create new category "Missing". Treat missingness as informative.

### D. Multivariate Imputation

- KNN Imputer: Use similar cases (neighbors) to estimate missing values. Uses relationships between variables.
- Iterative Imputer (MICE): Iteratively estimates missing values using other variables. Captures complex relationships.
- Regression Imputation: Predict missing values using regression on other variables. May underestimate variance.

## 5.5 Missing Data Techniques Summary

| Technique         | Best For    | When to Use               |
|-------------------|-------------|---------------------------|
| Listwise Deletion | MCAR        | Large sample, <5% missing |
| Pairwise Deletion | MCAR        | Different analyses needed |
| Mean/Median       | Numerical   | MCAR, univariate          |
| Mode              | Categorical | MAR, <5% missing          |
| Random Sample     | Both        | MCAR, >10% missing        |
| KNN Imputer       | Both        | Complex relationships     |
| Iterative/MICE    | Both        | MAR, multiple variables   |
| Arbitrary Value   | Both        | NMAR, informative missing |

## SECTION F: DATA VISUALIZATION

(Expected Marks: 5-8)

### 6.1 Tools for CATEGORICAL Data

- Frequency Distribution: Count of each category
- Relative Frequency: Proportion of each category
- Percent Frequency: Percentage of each category
- Bar Chart: Bars for each category (separated to emphasize distinctness)
- Pie Chart: Sectors proportional to frequencies

### Bar Charts vs Histograms

| Bar Chart                      | Histogram                     |
|--------------------------------|-------------------------------|
| For categorical data           | For quantitative data         |
| Bars are separated             | No separation between bars    |
| Emphasizes distinct categories | Shows continuous distribution |

### 6.2 Tools for QUANTITATIVE Data

- Dot Plot: Simplest summary. Dots above axis for each value.
- Histogram: Rectangles above class intervals. Height = frequency. No natural separation.
- Box Plot: Shows Min, Q1, Median, Q3, Max. Identifies outliers easily.

### 6.3 Summarizing Two Variables

Scatter Diagrams: Graphical presentation of relationship between TWO quantitative variables. One on horizontal, other on vertical. Pattern suggests overall relationship. Trendline provides approximation.

### 6.4 Best Practices

- Choose appropriate chart for data type
- Label axes clearly and use consistent scales
- Avoid chart junk (unnecessary decorations)
- Use color purposefully and include legends

## SECTION G: OUTLIER DETECTION

(Expected Marks: 5)

### 7.1 Definition

Observations that are distinctly different from the rest of the sample.

### 7.2 Classification of Outliers

- Procedural Error: Data entry or coding mistakes. Action: Delete or correct.
- Extraordinary Event: Valid but unique occurrences. Action: Retention depends on research goals.

### 7.3 Detection Methods

#### A. Visualization

- Scatter plots: Highlight points deviating from typical pattern
- Box plots: Points beyond whiskers are outliers

#### B. Statistical Methods (IMPORTANT)

1. Z-Score Method:

```
Z = (X - mean) / std_dev
Threshold: |Z| > 3 -> Outlier
```

Measures how far data point is from mean in standard deviations. Assumes approximately normal distribution.

2. Interquartile Range (IQR) Method:

```
IQR = Q3 - Q1
Lower Bound = Q1 - 1.5 * IQR
Upper Bound = Q3 + 1.5 * IQR
Any value outside [Lower, Upper] = Outlier
```

Advantage: Robust to outliers, no distribution assumption required.

### 7.4 Box Plot Components

- Box: Q1 to Q3 (IQR)
- Line inside box: Median
- Whiskers: Extend to min and max (excluding outliers)
- Points beyond whiskers: Outliers

## SECTION H: FEATURE ENGINEERING

(Expected Marks: 8-10)

### 8.1 Definition

Process of using domain knowledge to extract features from raw data to improve ML model performance. Relies heavily on domain expertise.

### 8.2 Feature Scaling (IMPORTANT)

**NOTE:**

Very common question! Understand all four scaling methods.

Purpose: Standardizing ranges so features with larger magnitudes do not unfairly dominate the model.

#### 1. Standardization (Z-score)

$$X_{\text{scaled}} = (X - \text{mean}) / \text{std\_dev}$$

Result: Mean = 0, Std Dev = 1

Use when: Data approximately normal, outliers present.

#### 2. Min-Max Scaling

$$X_{\text{scaled}} = (X - X_{\text{min}}) / (X_{\text{max}} - X_{\text{min}})$$

Result: Values between 0 and 1

Use when: Need bounded range, no outliers.

#### 3. Robust Scaling

$$X_{\text{scaled}} = (X - \text{median}) / \text{IQR}$$

Use when: Data with many outliers. Less sensitive to outliers.

#### 4. Normalizer

Scales each sample so feature vector has length of 1. Use for text classification and clustering.



## 8.3 Handling Categorical Values

### A. One-Hot Encoding (VERY IMPORTANT)

Convert categorical data into numerical by creating binary (0/1) columns for each category.

- Example: Animal column (Dog, Cat, Sheep) becomes 3 binary columns
- Use for: Unordered categories (nominal data)
- Caution: Creates many columns for high cardinality

### B. Label Encoding

Assign integer to each category. Use for target variable (Y) in classification. Caution: Implies order where none exists.

### C. Ordinal Encoding

Assign integers preserving order. Use for ordered categories (ordinal data). Example: Low=0, Medium=1, High=2.

## 8.4 Encoding Methods Summary

| Method           | Input X    | Output Y | When to Use          | Order? | Type        |
|------------------|------------|----------|----------------------|--------|-------------|
| Label Encoding   | Usually No | Yes      | Target classes       | No     | Target      |
| Ordinal Encoding | Yes        | Rarely   | Ordered categories   | Yes    | Categorical |
| One-Hot Encoding | Yes        | No       | Unordered categories | No     | Categorical |

## 8.5 Binning (Numerical to Categorical)

Grouping numerical subranges into bins/buckets. Turns numerical data into categorical. Types: Equal-width, Equal-frequency, Custom/domain-based.

## 8.6 Feature Construction

Creating new, highly informative variables from raw data. Manual or algorithmic. Relies on domain expertise. Example: Creating BMI from height and weight.

## 8.7 Feature Selection

- Filter methods: Based on statistical measures (correlation)
- Wrapper methods: Select based on model performance
- Embedded methods: Feature selection integrated within model

## 8.8 Feature Extraction

- Dimensionality Reduction: PCA reduces features while preserving variance
- Aggregation & Combination: Summing or averaging features

## 8.9 Principal Component Analysis (PCA)

Dimensionality reduction technique. Finds new features (principal components) that explain most variance. Process: Standardize -> Compute covariance -> Find eigenvectors -> Select top k -> Transform data.

## SECTION I: MACHINE LEARNING FUNDAMENTALS

(Expected Marks: 5-8)

### 9.1 Definition

Subfield of AI that enables systems to learn from data, improve performance on specific tasks through experience, without being explicitly programmed.

### 9.2 Components of ML

- Data: Training examples
- Model: Mathematical representation
- Algorithm: Learning procedure
- Loss Function: Measures prediction error
- Optimization: Minimizes loss function

### 9.3 Traditional Programming vs ML

| Aspect      | Traditional           | Machine Learning      |
|-------------|-----------------------|-----------------------|
| Input       | Rules + Data          | Data + Answers        |
| Output      | Answers               | Rules                 |
| Approach    | Explicitly code rules | Learn rules from data |
| Flexibility | Fixed behavior        | Adapts to new data    |

### 9.4 Types of Machine Learning

- Supervised Learning: Dataset includes labels. Types: Classification, Regression. Example: Titanic survival.
- Unsupervised Learning: No labels, find hidden patterns. Types: Clustering, Dimensionality Reduction. Example: Customer segmentation.
- Reinforcement Learning: Learning through rewards/penalties.

### 9.5 ML Workflow

```

Complete Dataset
|
Train Set -> Train Model
|
Validation Set -> Tune Model
|
Best Model Selected
|
Test Set -> Final Evaluation

```

### 9.6 Applications of ML

- Predictive analytics, Image and speech recognition
- Natural language processing, Recommendation systems
- Autonomous vehicles, Fraud detection
- Medical diagnosis, Financial forecasting

## SECTION J: CHALLENGES IN MACHINE LEARNING

(Expected Marks: 5-8)

### 10.1 Insufficient Quantity of Data

- Models require sufficient examples to capture generalizable patterns
- Small datasets -> low statistical power
- Risk of overfitting (memorizing noise instead of learning patterns)
- Solution: Data augmentation, transfer learning

### 10.2 Non-representative and Poor-Quality Data

**NOTE:**

80% of project time is consumed by data wrangling and cleaning!

- Issues: Incomplete, inaccurate, inconsistent, duplicate data
- Consequence: Model collapses on unseen data if training data is not representative
- Solution: Rigorous data validation and cleaning pipelines

### 10.3 Irrelevant Features

- Irrelevant variables bloat model complexity -> overfitting
- Multicollinearity: Correlated features mask true variable impact
- Solutions: Univariate filters, model-based ranking, wrapper methods

### 10.4 The Challenge of Estimation

Goal: Find optimal parameters that minimize loss. Challenge: Balance underfitting (high bias) vs overfitting (high variance). Sweet spot: Optimal model complexity.

### 10.5 Reducing Human Biases

- Data Bias: Algorithms absorb historical prejudices. Examples: Gender bias, occupational bias.
- Algorithmic Bias: Biases through programmer design choices. Proxy variables encode protected attributes.
- Solution: Fairness metrics built in from day one.

### 10.6 Ethical Use of Technology

- Privacy & Security: GDPR, CCPA, model inversion attacks
- Nefarious Applications: Deepfakes, adversarial attacks, disinformation
- Societal Impact: Workforce displacement, accountability, algorithmic decisions affecting credit/hiring/justice

### Challenges Summary

| Challenge           | Key Issue          | Solution                |
|---------------------|--------------------|-------------------------|
| Insufficient Data   | Overfitting        | Data augmentation       |
| Poor Quality Data   | Model collapse     | Cleaning, validation    |
| Irrelevant Features | Overfitting, noise | Feature selection       |
| Estimation          | Bias-variance      | Cross-validation        |
| Human Biases        | Unfair outcomes    | Fairness metrics        |
| Ethics              | Privacy, harm      | Regulations, frameworks |

## SECTION K: STATISTICAL CONCEPTS

(Expected Marks: 3-5)

### 11.1 Descriptive vs Inferential Statistics

- Descriptive: Description and summarization of data. Forms: Tabular, graphical, numerical. Examples: Mean, median, charts.
- Inferential: Drawing conclusions from data. Must account for probability. Examples: Hypothesis testing, confidence intervals.

### 11.2 Central Tendency Measures

| Measure | Best For                | Data Type                |
|---------|-------------------------|--------------------------|
| Mean    | Symmetric distributions | Interval, Ratio          |
| Median  | Skewed, outliers        | Ordinal, Interval, Ratio |
| Mode    | Categorical data        | Nominal, Ordinal         |

### 11.3 Dispersion Measures

- Range: Max - Min
- Variance: Average squared deviation from mean
- Standard Deviation: Square root of variance
- IQR: Q3 - Q1 (robust to outliers)

## SECTION L: LEARNING TYPES

(Expected Marks: 3)

### 12.1 Supervised Learning

Dataset includes labels (correct answers). Goal: Learn mapping from inputs to outputs. Types: Classification (discrete categories) and Regression (continuous values). Example: Titanic survival prediction.

### 12.2 Unsupervised Learning

No labels; machine finds hidden patterns. Goal: Discover structure in data. Types: Clustering (group similar points) and Dimensionality Reduction. Example: Customer segmentation.

### 12.3 Key Differences

| Aspect     | Supervised                 | Unsupervised      |
|------------|----------------------------|-------------------|
| Data       | Labeled                    | Unlabeled         |
| Goal       | Prediction                 | Pattern discovery |
| Evaluation | Clear metrics              | Subjective        |
| Examples   | Classification, Regression | Clustering, PCA   |

## APPENDIX A: IMPORTANT FORMULAS

Memorize these formulas for the exam:

|                                  |                                                        |
|----------------------------------|--------------------------------------------------------|
| <b>Z-Score (Standard Score):</b> | $Z = (X - \text{mean}) / \text{std\_dev}$              |
| <b>Interquartile Range:</b>      | $IQR = Q3 - Q1$                                        |
| <b>Outlier Lower Bound:</b>      | $Q1 - 1.5 * IQR$                                       |
| <b>Outlier Upper Bound:</b>      | $Q3 + 1.5 * IQR$                                       |
| <b>Standardization:</b>          | $X\_scaled = (X - \text{mean}) / \text{std\_dev}$      |
| <b>Min-Max Scaling:</b>          | $X\_scaled = (X - X\_min) / (X\_max - X\_min)$         |
| <b>Mean:</b>                     | $\text{mean} = \text{sum}(X) / n$                      |
| <b>Median:</b>                   | Middle value (or avg of two middle values)             |
| <b>Mode:</b>                     | Most frequently occurring value                        |
| <b>Variance:</b>                 | $\text{sigma}^2 = \text{sum}((X - \text{mean})^2) / n$ |
| <b>Standard Deviation:</b>       | $\text{sigma} = \text{sqrt}(\text{variance})$          |

## APPENDIX B: EXAM PREPARATION TIPS

### Expected Question Pattern

5 Questions total: 3 Compulsory + 2 Optional (out of given options)

### Most Likely Compulsory Questions

**Q1: AI Definitions + Data Types (Nominal, Ordinal, Interval, Ratio)**

**Expected: 8-10 marks**

**Q2: EDA Steps + Missing Data (MCAR, MAR, NMAR) + Imputation**

**Expected: 10-12 marks**

**Q3: Feature Engineering (Scaling, Encoding, Outlier Detection)**

**Expected: 10-12 marks**

### Likely Optional Questions

- Option A: Knowledge Representation + Data Quality Dimensions (8-10 marks)
- Option B: ML Challenges + Statistical Concepts + Learning Types (8-10 marks)

### Key Diagrams to Practice Drawing

- ML Pipeline / Workflow diagram
- Missing Data Handling flowchart (Complete Removal vs Imputation)
- Box Plot with outlier detection labels
- One-Hot Encoding transformation example
- PCA dimension visualization
- Data Types hierarchy (Nominal -> Ordinal -> Interval -> Ratio)

### Key Definitions to Memorize

- AI, Intelligence, Turing Test
- MCAR, MAR, NMAR with examples
- EDA and its 7 steps
- Feature Scaling: Standardization, Min-Max, Robust, Normalizer
- One-Hot Encoding, Label Encoding, Ordinal Encoding
- Outlier detection: Z-score and IQR methods
- 8 Dimensions of Data Quality
- 6 Major Challenges in ML

### Python Code Snippets (if asked)

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

df = pd.read_csv('data.csv')
df.shape()          # Rows and columns
df.info()           # Data types, missing values
df.describe().T     # Statistical summary
df.isnull().sum()   # Missing values count
```

**BEST OF LUCK FOR YOUR MID-SEM EXAM!**

*Prepare well. You have got this!*